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NoZZoMe: An Open-Source Parametric Geometry Framework for MOC-Assisted Genetic Optimization of Single Bell Rocket Nozzles

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Software

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Summary

A rocket nozzle converts the thermal and pressure energy of combustion gases into a high-speed exhaust jet. Its contour affects thrust, efficiency, length, and the risk of undesirable flow behaviour, but exploring alternative contours commonly requires several disconnected tools. NoZZoMe is an open-source Python application for generating, analysing, and optimizing axisymmetric single-bell rocket-nozzle geometries. It combines thermochemical properties, parametric geometry, reduced-order flow and loss models, a Method of Characteristics (MOC) analysis, genetic optimization, interactive visualization, and machine-readable export in one reproducible workflow.

NoZZoMe is intended for researchers, student rocketry teams, and preliminary-design engineers who need to inspect the assumptions and intermediate results of a nozzle study. Users can operate it through a desktop interface or a shared Python simulation API. The software is deliberately positioned between ideal sizing calculations and mesh-refined computational fluid dynamics (CFD): it supports rapid and transparent design comparison but is not a qualification tool.

Statement of need

Preliminary nozzle design involves more than drawing a divergent wall. An operating point must be connected to thermochemical properties and an expansion ratio; a geometrically continuous contour must then be evaluated for inviscid expansion, multidimensional turning, boundary-layer displacement, wall friction, and performance losses. These calculations are often split across thermochemical programs, spreadsheets, contour utilities, optimization scripts, and CFD packages. This fragmentation makes assumptions difficult to trace and encourages inconsistent evaluation paths between manually generated and optimized designs.

NoZZoMe addresses this gap with an inspectable workflow in which the graphical interface, examples, and optimizers use the same maintained simulation components. It supports pressure-matched expansion sizing, continuous convergent-throat-bell geometry, quasi-one-dimensional profiles, adiabatic thermal diagnostics, BLIMP-inspired boundary-layer closures, prescribed-wall axisymmetric MOC analysis, and genetic optimization. The resulting contours and physical fields can be exported as CSV and JSON for independent analysis, CAD preparation, and reproducible research. The validity limits of every reduced-order model are documented, and final designs remain subject to CFD, structural and thermal assessment, manufacturing review, and experimental validation.

State of the field

NoZZoMe occupies a different level of scope and fidelity from established propulsion, optimization, and CFD tools. Its closest relationships are summarized below.

Table 1: Positioning of NoZZoMe relative to complementary engineering software.

Software	Primary role	Relationship to NoZZoMe
RPA (RP Software+Engineering UG, n.d.)	Broad preliminary chemical-rocket-engine analysis, including performance, chamber, nozzle, and thermal calculations	Broader engine-level scope; NoZZoMe concentrates on an inspectable Python workflow for single-bell contour generation and optimization
RocketCEA (DeLong, n.d.; Gordon & McBride, 1994)	Python access to NASA CEA thermochemistry and ideal rocket performance	Used by NoZZoMe as a backend; NoZZoMe adds continuous geometry, prescribed-wall MOC, viscous corrections, and design search
OpenMDAO (Gray et al., 2019)	General multidisciplinary analysis and optimization architecture	A general orchestration framework rather than a nozzle-specific modelling application
SU2 (Economon et al., 2016)	Mesh-based multiphysics simulation and PDE-constrained design	A higher-fidelity and higher-cost downstream option for checking shortlisted NoZZoMe contours

Contributing the complete workflow to any one of these projects would mix a specialized nozzle-design application with either a thermochemical backend, a general workflow framework, or a mesh-based CFD suite. NoZZoMe instead composes established packages where appropriate and contributes the nozzle-specific geometry, physical closures, verification logic, interactive workflow, and exact-finalist optimization strategy. It complements rather than replaces the cited tools.

Software design

The overall analysis and optimization sequence is summarized in Figure 1.

Shared physical evaluation path

The central architectural decision is to keep one physical evaluation path across interactive simulation and optimization. Operating conditions, engine pre-sizing quantities, fixed contour inputs, and genetic variables are represented separately so that the optimizer cannot silently change quantities that should remain fixed. The maintained Python packages separate geometry, RocketCEA access, flow reconstruction, thermal diagnostics, boundary-layer integration, performance, MOC, optimization, and export, while a shared simulation entry point coordinates the common reduced-order analysis.

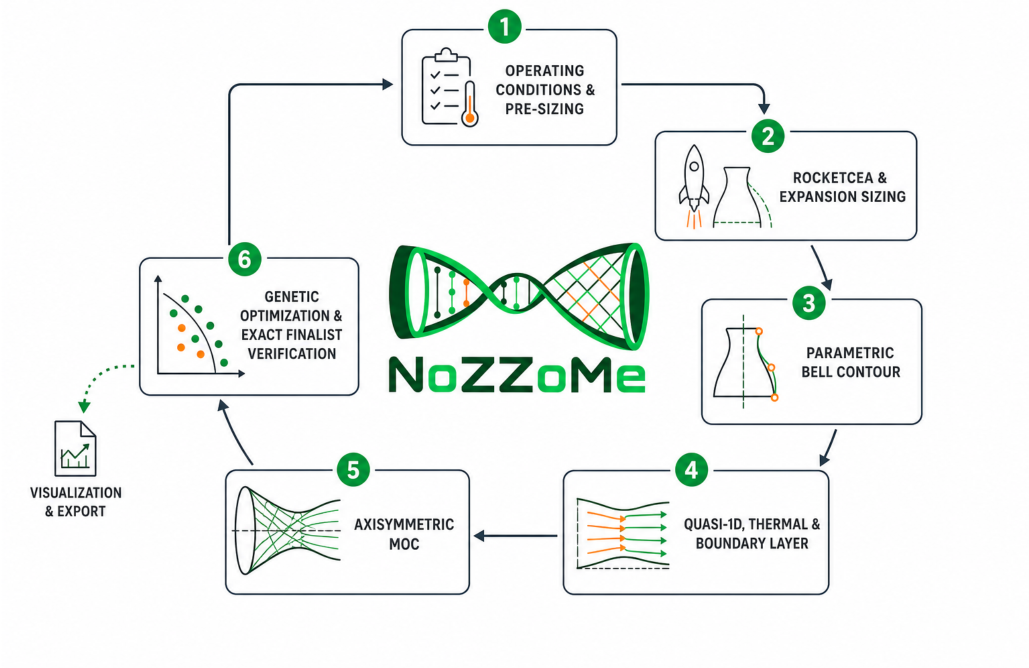

Figure 1: NoZZoMe analysis and optimization workflow. Reduced-order and axisymmetric models support the genetic search, while shortlisted designs undergo exact finalist verification before visualization and export.

Modelling and analysis capabilities

Table 2: Main technical capabilities exposed by the common NoZZoMe workflow.

Capability	Technical implementation	Principal output
Thermochemistry and sizing	RocketCEA properties with pressure-matched expansion sizing	Throat and exit conditions, expansion ratio, ideal performance
Parametric geometry	Continuous convergent, throat, and single-bell contour construction	Wall coordinates and geometric design variables
Reduced-order analysis	Quasi-one-dimensional flow, adiabatic thermal diagnostics, wall friction, and integral boundary-layer models	Axial profiles, displacement effects, and loss estimates
Axisymmetric MOC	Prescribed-wall characteristic marching with characteristic-field visualization	Mach, pressure, flow-angle, and mass-conservation diagnostics
Design optimization	Genetic search with Quick, BLIMP-lite, or MOC-assisted evaluation	Ranked feasible geometries and an exactly verified finalist
Reproducibility	Shared API, desktop interface, automated tests, and CSV/JSON export	Repeatable studies and machine-readable results

Multi-fidelity optimization and verification

Three optimization fidelities balance computational cost and physical detail. Quick uses an integral boundary-layer closure, BLIMP-lite uses a resolved compressible profile marcher, and MOC-assisted mode includes multidimensional turning. Because refined MOC is too costly for every genetic candidate, a bounded piecewise-linear surrogate is constructed from exact MOC evaluations. Shortlisted candidates are then recomputed with exact coarse MOC, finalists on Fast or Precise meshes, and only an exactly verified design is returned.

Verification prevents surrogate acceleration from concealing physical error. MOC solutions are checked against the RocketCEA choking reference and for mass conservation. Invalid geometry, non-finite states, predicted separation, or excessive residuals are rejected. Numerical studies, retained CSV results, automated tests, and continuous integration provide reproducible checks beyond the graphical interface.

Interactive inspection

Representative outputs are shown in Figure 2 and Figure 3. They expose the generated nozzle geometry and the axisymmetric characteristic-field diagnostics.

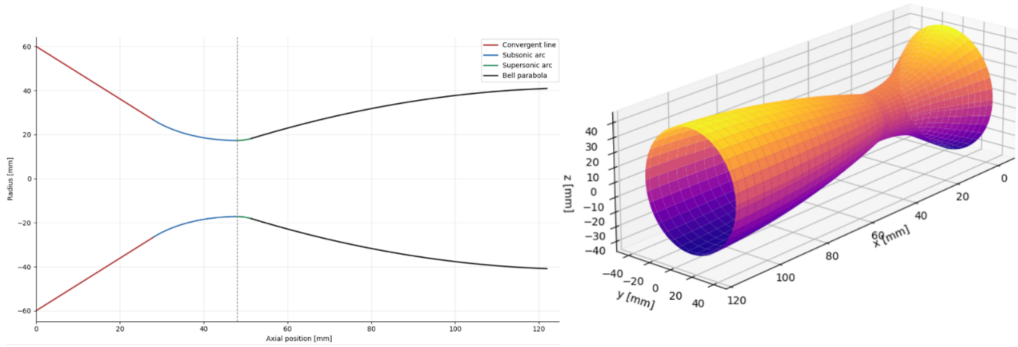

Figure 2: Parametric single-bell nozzle geometry: two-dimensional contour construction and interactive three-dimensional surface visualization.

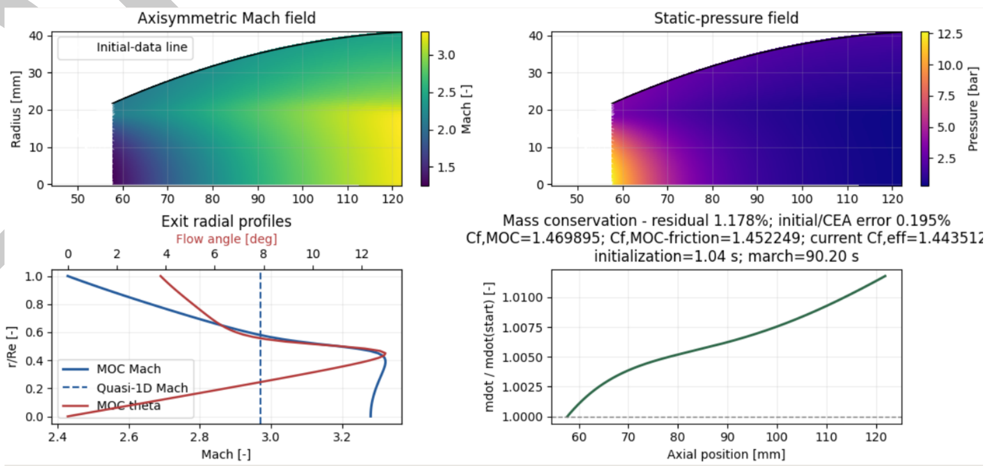

Figure 3: Axisymmetric MOC diagnostics: Mach-number and static-pressure fields, exit radial profiles, and mass-conservation verification.

Research impact statement

NoZZoMe originated in the Porto Space Team workflow for preliminary analysis of the paraffin/nitrous-oxide hybrid propulsion system developed in the context of the European Rocketry Challenge. Its evolution is publicly documented in a repository with development activity since June 2025 and contributions from several team members. Release v0.2.0 provides the first formal research-software snapshot (Lino & Ferreira, 2026a).

The accompanying technical report documents the underlying geometric formulation, physical models, verification procedures, and MOC-assisted optimization study (Lino & Ferreira, 2026b). Reproducible scripts in the repository regenerate the principal model comparisons and convergence studies. In the reported reference case, the Fast and Precise finalist meshes selected the same geometry, while the refined study quantified the remaining performance difference and enforced independent mass-flow thresholds. These materials demonstrate present research use by the developers and provide auditable benchmarks for reuse, extension, and comparison by other propulsion researchers. The public API, contribution guidelines, issue tracker, GPL-3.0 license, tests, and machine-readable outputs are intended to support such use beyond the original team.

AI usage disclosure

OpenAI Codex, using GPT-5-family models available in August 2026, assisted with code refactoring, test scaffolding, documentation organization, editorial review, and drafting and copy-editing portions of this paper and the related technical report. The human authors defined the research problem, selected the physical and numerical models, made the architectural and scientific decisions, reviewed and edited all AI-assisted material, and validated software behaviour and reported numerical results against the documented equations, convergence studies, automated tests, and retained outputs. The authors remain responsible for the accuracy, originality, licensing, and scientific claims of the software and publications.

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